



BAULKHAM HILLS HIGH SCHOOL

2016
YEAR 12 TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 6 – 12

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

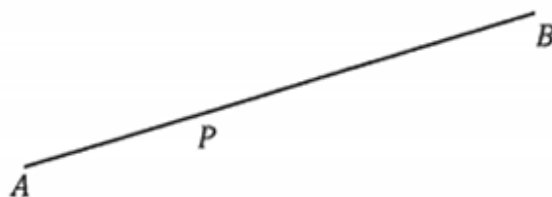
- 1 Suppose θ is the acute angle between the lines $y - 2x = 3$ and $3y = -x + 2$. Which of the following is the value of $\tan\theta$?

- (A) 7
- (B) -7
- (C) 1
- (D) -1

- 2 Which of the following is an expression for $\int \sin^2 6x \, dx$?

- (A) $\frac{x}{2} - \frac{1}{12} \sin 6x + c$
- (B) $\frac{x}{2} + \frac{1}{12} \sin 6x + c$
- (C) $\frac{x}{2} - \frac{1}{24} \sin 12x + c$
- (D) $\frac{x}{2} + \frac{1}{24} \sin 12x + c$

- 3 The point P divides the interval AB in the ratio 3:8



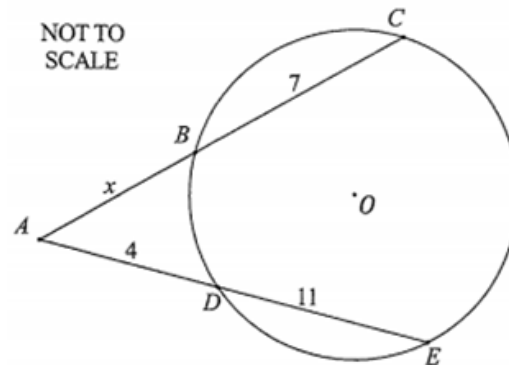
In what external ratio does A divide the interval PB ?

- (A) 11:3
- (B) 8:3
- (C) 3:5
- (D) 3:11

- 4 The expression $\sin x - \sqrt{3} \cos x$ can be written in the form $2\sin(x + \alpha)$.
Find the value of α .

- (A) $\frac{\pi}{3}$
(B) $\frac{\pi}{6}$
(C) $-\frac{\pi}{3}$
(D) $-\frac{\pi}{6}$

- 5 In the diagram below, BC and DE are chords of a circle. CB and ED are produced to meet at A .



What is the value of x ?

- (A) 12
(B) 5
(C) $\frac{28}{11}$
(D) $\frac{11}{28}$
- 6 Four female and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate?
- (A) $4! \times 4!$
(B) $3! \times 4!$
(C) $3! \times 3!$
(D) $2 \times 3! \times 4!$

- 7 It is known that $(x + 2)$ is a factor of the polynomial $P(x)$ and that

$$P(x) = (x^2 + x + 1) \times Q(x) + (2x + 3)$$

for some polynomial $Q(x)$. From this information alone, it can be deduced that;

(A) $Q(-2) = \frac{1}{3}$

(B) $Q(-2) = -\frac{1}{3}$

(C) $Q(-2) = 1$

(D) $Q(-2) = -1$

- 8 The velocity of a particle moving in simple harmonic motion in a straight line is given by $v^2 = 2 - x - x^2 \text{ ms}^{-1}$, where x is displacement in metres.

Find the centre of the motion.

(A) $x = 2$

(B) $x = 1$

(C) $x = -\frac{1}{2}$

(D) $x = -2$

- 9 How many solutions does the equation $2x + 3\pi \sin x = 0$ have in the domain $0 \leq x \leq 2\pi$?

(A) 1

(B) 2

(C) 3

(D) 4

10 Which one of the following is the general solution of $2\sin^2\left(6t + \frac{\pi}{4}\right) = 1$?

(A) $t = \frac{n\pi}{12}$, where n is an integer.

(B) $t = \frac{n\pi}{12} - \frac{\pi}{24}$, where n is an integer.

(C) $t = \frac{n\pi}{3}$ and $t = \frac{n\pi}{3} + \frac{\pi}{12}$, where n is an integer.

(D) $t = \frac{n\pi}{3} - \frac{\pi}{6}$ and $t = \frac{n\pi}{3} + \frac{\pi}{12}$, where n is an integer.

END OF SECTION I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour 45 minutes for this section

Answer each question on the appropriate answer sheet. Each answer sheet must show your BOS#. Extra paper is available.

In Questions 11 to 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a *separate* answer sheet

Marks

(a) Solve the inequality $\frac{1-x}{1+x} \leq 1$ 3

- (b) Dong and Kevin are two of ten candidates for a committee. 2
How many ways can a committee of five be chosen, if Kevin refuses to be on the same committee as Dong?

(c) (i) Show that $\frac{\sin 2x}{1 + \cos 2x} = \tan x$ 2

(ii) Hence show that $\tan 15^\circ + \cot 15^\circ = 4$ 2

(d) Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} dx$ using the substitution $x = u^2 - 1$ 3

- (e) A particle moves in a straight line along the x axis so that its acceleration is given by;

$$\ddot{x} = \frac{1}{4+x^2}$$

where x is the displacement from the origin.

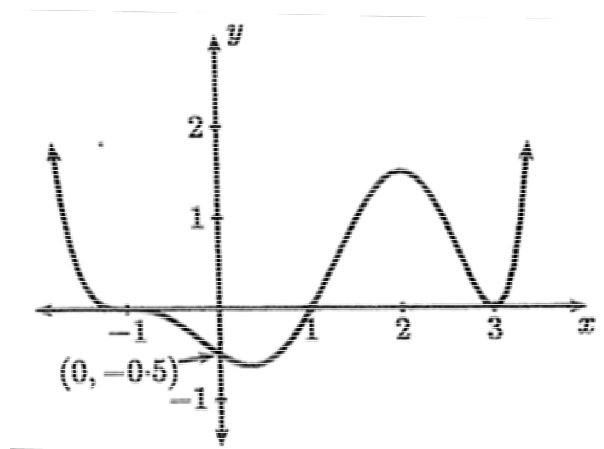
Initially the particle is at rest at the origin.

- (i) Find v^2 as a function of x . 2
(ii) Explain why v is always positive for $t > 0$ 1

Question 12 (15 marks) Use a *separate* answer sheet

(a)

2



Write down a possible function $y = P(x)$ for the polynomial function sketched above. (Do NOT use calculus)

(b) Find $\int \frac{dx}{\sqrt{16 - 9x^2}}$

2

(c) Water is being heated in a kettle. At time t seconds, the temperature of the water is $T^\circ \text{C}$.

The rate of increase of the temperature of the water at any time after the kettle is switched on, is modelled by the equation $\frac{dT}{dt} = k(120 - T)$, where k is a positive constant.

The temperature of the water is at 20°C when the kettle is switched on.

- (i) Show that $T = 120 - 100e^{-kt}$ is both a solution to the differential equation and satisfies the initial conditions. 2
- (ii) When the temperature of the water reaches 100°C , the kettle switches off. If it takes 10 seconds for the temperature to increase 10°C , once the kettle is switched on, find how long it takes for the kettle to switch off, to the nearest second. 2

Question 12 continues on page 8

Question 12 (continued)

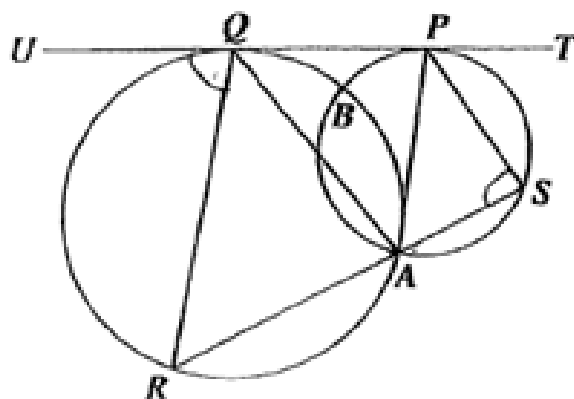
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|---------|--|---|
| (d) (i) | Sketch the graph of the function $f(x) = e^x - 4$ showing clearly the coordinates of any points of intersection with the axes and the equations of any asymptotes. | 1 |
| (ii) | On the same diagram sketch the graph of the function $y = f^{-1}(x)$ showing clearly the coordinates of any points of intersection with the axes and the equations of any asymptotes. | 1 |
| (iii) | Find an expression for $y = f^{-1}(x)$ in terms of x | 2 |
| (iv) | Explain why the x coordinate of any point of intersections of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $e^x - x - 4 = 0$ | 1 |
| (v) | Taking $x_1 = -4$ as a first approximation, use one application of Newton's Method to find a second approximation for the point of intersection located in the third quadrant. Give your answer correct to two decimal places. | 2 |

End of Question 12

Question 13 (15 marks) Use a *separate* answer sheet

- (a) Use mathematical induction to prove that $3^{2n+4} - 2^{2n}$ is divisible by 5, for $n \geq 1$ 3

- (b) In the diagram below, two circles intersect at A and B . The common tangent TU touches the circles at P and Q respectively. A line through A cuts the left hand circle at R and the right hand circle at S , and it is found that $PQRS$ is a cyclic quadrilateral.



Copy the diagram into your answer booklet.

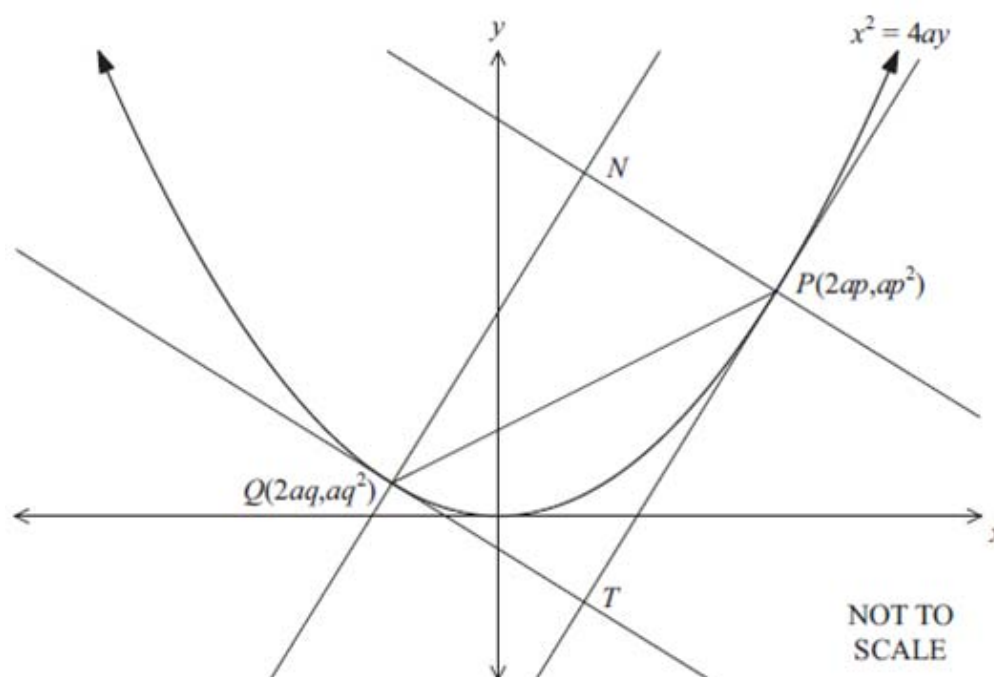
- | | |
|--|---|
| (i) Give a reason why $\angle UQR = \angle PSA$ | 1 |
| (ii) Prove that $PS \parallel AQ$. | 2 |
| (iii) Thus show that $\triangle PAS \parallel \triangle QRA$ | 2 |

Question 13 continues on page 10

Question 13 (continued)

- (c) The diagram shows the parabola $x^2 = 4ay$. The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola such that PQ is a focal chord.

The tangents at P and Q intersect at T and the normal at P and Q intersect at N .



- | | |
|---|---|
| (i) Show that the equation of any chord PQ is given by $(p + q)x - 2y = 2apq$ | 2 |
| (ii) Show that $pq = -1$ | 1 |
| (iii) Explain why $PTQN$ is a cyclic quadrilateral | 1 |
| (iv) Let C be the centre of the circle that passes through the points P, T, Q and N . Explain why C is the midpoint of PQ . | 1 |
| (v) Find the Cartesian equation of the locus of C . | 2 |

End of Question 13

Question 14 (15 marks) Use a *separate* answer sheet

- (a) (i) Neatly sketch the graph of $y = \sin^{-1}\left(\frac{x}{2}\right)$ clearly indicating the domain and range. 2

- (ii) By considering the graph in part (i), find the exact value of; 2

$$\int_0^1 \sin^{-1}\left(\frac{x}{2}\right) dx$$

- (b) A particle is moving in a straight line according to the equation;

$$x = 4\sin^2 t$$

where x is the displacement in metres and t is the time in seconds.

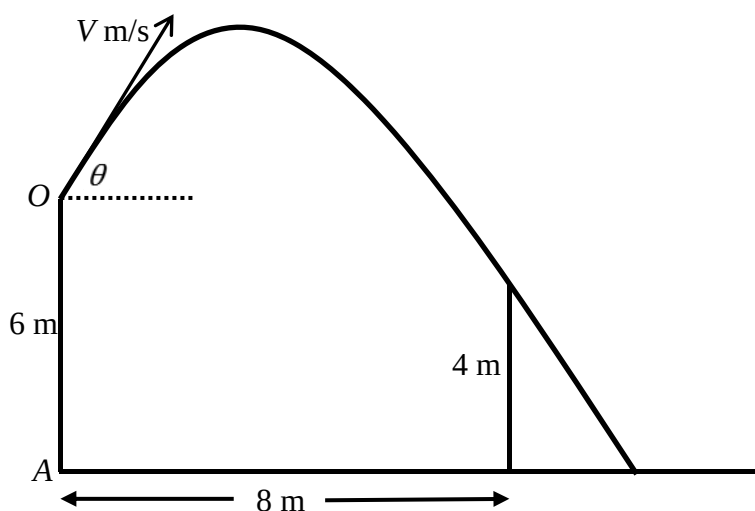
- (i) Show that the particle moves in Simple Harmonic Motion. 2
- (ii) Find the value of x for which the speed is a maximum and determine that speed. 2

Question 14 continues on page 12

Question 14 (continued)

- (c) A projectile is fired from a point O , which is 6 metres above horizontal ground, with initial velocity V m/s, at an angle of θ to the horizontal.

There is a thin vertical post which is 4 metres high and 8 metres horizontally away from a point A , directly below O , as shown in the diagram below.



The equations of motion are given by;

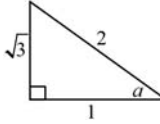
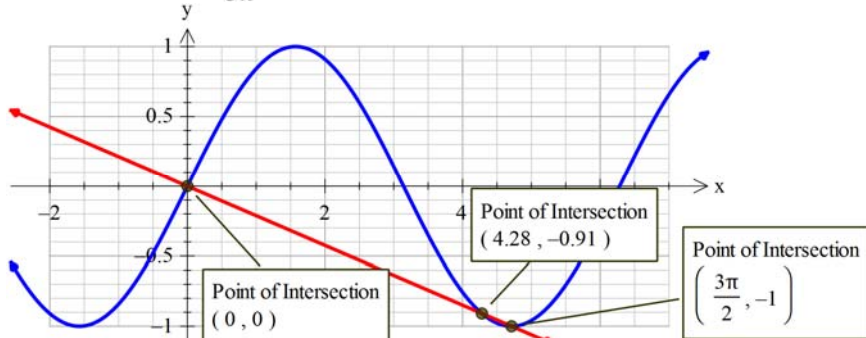
$$x = Vt\cos\theta \quad \text{and} \quad y = Vt\sin\theta - 4.9t^2$$

(Do NOT prove this)

- | | | |
|-------|---|---|
| (i) | If 2 seconds after projection, the projectile passes just above the top of the post, show that $\tan\theta = 2.2$ | 2 |
| (ii) | Show that the projectile hits the ground approximately 0.3 seconds after passing over the post. | 3 |
| (iii) | Find the angle that the projectile makes with the ground when it hits the ground, correct to the nearest degree. | 2 |

End of paper

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YEAR 12 EXTENSION 1 TRIAL 2016 SOLUTIONS

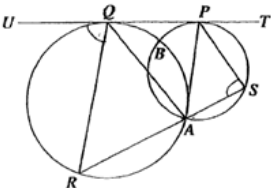
Solution	Marks	Comments
SECTION I		
1. A – $y - 2x = 3 \Rightarrow m_1 = 2$ $3y = -x + 2 \Rightarrow m_2 = -\frac{1}{3}$ $\tan \theta = \left \frac{2 + \frac{1}{3}}{1 + (2)\left(-\frac{1}{3}\right)} \right $ $= \left \frac{\frac{7}{3}}{\frac{1}{3}} \right $ $= 7$	1	
2. C – $\int \sin^2 6x dx = \frac{1}{2} \int (1 - \cos 12x) dx$ $= \frac{1}{2} \left(x - \frac{1}{12} \sin 12x \right) + c$ $= \frac{x}{2} - \frac{1}{24} \sin 12x + c$	1	
3. D – $AP = 3x$, $BP = 8x$ $\therefore \frac{AP}{AB} = \frac{3x}{11x}$ $= \frac{3}{11}$	1	
4. C – $\sin x - \sqrt{3} \cos x = 2 \sin \left(x - \frac{\pi}{3} \right)$ $= 2 \sin(x + \alpha)$ $\therefore \alpha = -\frac{\pi}{3}$ $\tan \alpha = \sqrt{3}$ $\alpha = \frac{\pi}{3}$ 	1	
5. B – $AB \times AC = AD \times AE$ (product of intercepts of intersecting secants) $x(x+7) = 4 \times 15$ $x^2 + 7x - 60 = 0$ $(x+12)(x-5) = 0$ $x = -12 \text{ or } x = 5 \quad \text{but } x > 0 \therefore x = 5$	1	
6. B – Ways = $3! \times 4!$	1	
7. A – $P(-2) = 0$ $\left[(-2)^2 - 2 + 1 \right] Q(-2) + [2(-2) + 3] = 0$ $3Q(-2) - 1 = 0$ $Q(-2) = \frac{1}{3}$	1	
8. C – $v^2 \geq 0$ $2 - x - x^2 \geq 0$ $x^2 + x - 2 \leq 0$ $(x+2)(x-1) \leq 0$ $-2 \leq x \leq 1$ $\therefore$ centre is $x = -\frac{1}{2}$	1	
9. C – $2x + 3\pi \sin x = 0$ From the graph, there are three points of intersection $\sin x = -\frac{2}{3\pi} x$ 	1	

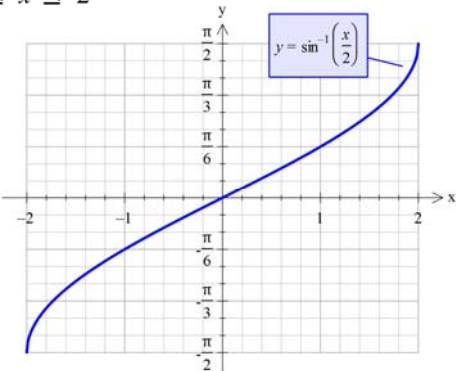
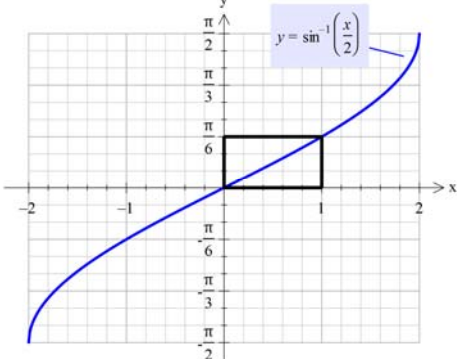
Solution	Marks	Comments
10. A - $2\sin^2\left(6t + \frac{\pi}{4}\right) = 1$ $1 - 2\sin^2\left(6t + \frac{\pi}{4}\right) = 0$ $\cos\left(12t + \frac{\pi}{2}\right) = 0$ $-\cos\left(\frac{\pi}{2} - 12t\right) = 0$ $\sin 12t = 0$ $12t = n\pi + (-1)^n \sin^{-1} 0 \text{ where } n \text{ is an integer}$ $12t = n\pi$ $t = \frac{n\pi}{12}$	1	

SECTION II		
Solution	Marks	Comments
QUESTION 11		
11(a) $\frac{1-x}{1+x} \leq 1$ $1+x \neq 0$ $x \neq -1$ $1-x = 1+x$ $2x = 0$ $x = 0$  $x < -1 \text{ or } x \geq 0$	3	3 marks - Correct graphical solution on number line or algebraic solution, with correct working 2 marks - Bald answer - Identifies the two correct critical points via a correct method 1 mark - Correct conclusion to their critical points obtained using a correct method 0 marks - Solves like a normal equation, with no consideration of the denominator.
11(b) Committees = ${}^{10}C_5 - {}^8C_5$ $= 252 - 56$ $= 196$	2	2 marks - Correct solution 1 mark - Identifies logical cases that must be considered - Successfully finds one of the required cases
11(c) (i) $\frac{\sin 2x}{1 + \cos 2x} = \frac{2\sin x \cos x}{1 + 2\cos^2 x - 1}$ $= \frac{2\sin x \cos x}{2\cos^2 x}$ $= \frac{\sin x}{\cos x}$ $= \tan x$	2	2 marks - Correct solution 1 mark - Correctly uses both $\sin 2\theta$ and $\cos 2\theta$ results
11 (c) (ii) $\tan 15^\circ = \frac{\sin 30^\circ}{1 + \frac{1}{2} \cos 30^\circ}$ $= \frac{\frac{1}{2}}{1 + \frac{\sqrt{3}}{2}}$ $= \frac{1}{2 + \sqrt{3}}$ $\tan 15^\circ + \cot 15^\circ = \frac{1}{2 + \sqrt{3}} + \frac{2 + \sqrt{3}}{1}$ $= \frac{1 + (2 + \sqrt{3})^2}{2 + \sqrt{3}}$ $= \frac{2 + \sqrt{3} + 4 + 4\sqrt{3} + 3}{2 + \sqrt{3}}$ $= \frac{8 + 4\sqrt{3}}{2 + \sqrt{3}}$ $= 4$	2	2 marks - Correct solution 1 mark - Uses given result to find $\tan 15^\circ$

Solution		Marks	Comments
11 (d)	$\int_0^3 \frac{x}{\sqrt{x+1}} dx = \int_1^2 \frac{u^2-1}{u} \times 2u du$ $= 2 \int_1^2 (u^2-1) du$ $= 2 \left[\frac{1}{3} u^3 - u \right]_1^2$ $= 2 \left(\frac{8}{3} - 2 - \frac{1}{3} + 1 \right)$ $= \frac{8}{3}$ $x = u^2 - 1 \quad \text{when } x = 0, u = 1$ $dx = 2u du \quad \text{when } x = 3, u = 2$	3	3 marks • Correct solution using the given substitution • <i>Note: solving as an indefinite integral, then using answer to find definite integral is acceptable</i> 2 marks • Correct primitive in terms of u • Correct integrand in terms of u, including the correct limits 1 mark • Correct integrand in terms of u without the limits • Correctly finds answer using an alternative approach
11 (e) (i)	$v \frac{dv}{dx} = \frac{1}{4+x^2}$ $\int_0^v v dv = \int_0^x \frac{dx}{4+x^2}$ $\frac{1}{2} [v^2]_0^v = \frac{1}{2} \left[\tan^{-1} \left(\frac{x}{2} \right) \right]_0^x$ $v^2 = \tan^{-1} \left(\frac{x}{2} \right)$	2	2 marks • Correct solution 1 mark • Correctly uses the idea that $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ or equivalent
11 (e) (ii)	$\ddot{x} = \frac{1}{4+x^2} > 0 \text{ for all values of } x$ Since particle begins at rest, positive acceleration means the particle moves in the positive direction, and will not slow down unless acceleration becomes negative. As this never occurs, $v > 0$	1	1 mark • Correct explanation
QUESTION 12			
12 (a)	By examining the zeros in the sketch, the polynomial must take the form $y = k(x+1)^3(x-1)(x-3)^2$ y intercept is $-\frac{1}{2}$ $\therefore a(1)(-1)(9) = -\frac{1}{2}$ $-9a = -\frac{1}{2}$ $a = \frac{1}{18} \quad \therefore \text{a possible polynomial is } y = \frac{1}{18}(x+1)^3(x-1)(x-3)^2$	2	2 marks • Correct solution 1 mark • Uses y intercept in a valid attempt to find function • Identifies the nature of the zeros
12(b)	$\int \frac{dx}{\sqrt{16-9x^2}} = \frac{1}{3} \int \frac{dx}{\sqrt{\frac{16}{9}-x^2}}$ $= \frac{1}{3} \sin^{-1} \left(\frac{x}{\frac{4}{3}} \right) + c$ $= \frac{1}{3} \sin^{-1} \frac{3x}{4} + c$	2	2 marks • Correct solution 1 mark • Uses correct standard integral
12 (c) (i)	$T = 120 - 100e^{-kt}$ $\frac{dT}{dt} = 100ke^{-kt}$ $= k(100e^{-kt} - 120 + 120)$ $= k(120 - T)$ $\text{when } t = 0, T = 120 - 100e^0 = 120 - 100 = 20$	2	2 marks • Correct solution 1 mark • Establishes initial temperature is 20° • Verifies given equation is solution to the differential equation

Solution	Marks	Comments
12 (c) (ii) when $t = 10$ $T = 30$ when $T = 100$, $100 = 120 - 100e^{-kt}$ $100e^{-kt} = 20$ $e^{-kt} = \frac{1}{5}$ $-kt = \ln \frac{1}{5}$ $t = -\frac{1}{k} \ln \frac{1}{5}$ $t = 152.7553185...$ The kettle switches off after 153 seconds when $t = 10$ $T = 30$ $30 = 120 - 100e^{-10k}$ $e^{-10k} = \frac{9}{10}$ $-10k = \ln \frac{9}{10}$ $k = -\frac{1}{10} \ln \frac{9}{10}$ $= 0.01053605157...$	2	2 marks • Correct solution 1 mark • Finds correct value of k • Uses their value of k correctly to find a solution <i>Note: no rounding penalty</i>
12 (d) (i) and (ii)	1	(i) 1 mark • Correct graph with all features labelled
12 (d) (iii) $f^{-1} : x = e^y - 4$ $e^y = x + 4$ $y = \ln(x + 4)$	2	2 marks • Correct solution 1 mark • Finds an expression in terms of y
12 (d) (iv) The two curves intersect on the line $y = x$ $\therefore x = e^x - 4$ $e^x - x - 4 = 0$	1	1 mark • Correct explanation
12 (d) (v) $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $= x_0 - \frac{e^x - x - 4}{e^x - 1}$ $= -4 - \frac{e^{-4} + 4 - 4}{e^{-4} - 1}$ $= -3.98134...$ $= -3.98 \text{ (correct to 3 decimal places)}$	2	2 marks • Correct solution 1 mark • Uses Newton's Method correctly

Solution	Marks	Comments
QUESTION 13		
13 (a) When $n = 1$; $3^{(2(1) + 4)} - 2^{2(1)} = 3^6 - 2^2$ $= 729 - 4$ $= 725 \quad \text{which is divisible by 5}$ Hence the result is true for $n = 1$ Assume the result is true for $n = k$ i.e. $3^{2k+4} - 2^{2k} = 5P$ where P is an integer Prove the result is true for $n = k + 1$ i.e. Prove $3^{2k+6} - 2^{2k+2} = 5Q$ where Q is an integer PROOF: $3^{2k+6} - 2^{2k+2} = 3^2(3^{2k+4}) - 2^{2k+2}$ $= 9(5P + 2^{2k}) - 2^2(2^{2k}) \quad (\text{by assumption})$ $= 45P + 9 \times 2^{2k} - 4 \times 2^{2k}$ $= 45P + 5 \times 2^{2k}$ $= 5(9P + 2^{2k})$ $= 5Q \quad \text{where } Q = 9P + 2^{2k} \text{ which is an integer}$ Hence the result is true for $n = k + 1$, if it is true for $n = k$ Since the result is true for $n = 1$, then it is true for all positive integers by induction.	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 mark • Successfully does 2 of the 4 key parts
13 (b) (i) exterior angle of a cyclic quadrilateral equals the opposite interior angle	1	1 mark • Correct reason
13 (b) (ii) $\angle UQR = \angle QAR$ (alternate segment theorem) $\therefore \angle PSA = \angle QAR$ (explained in (i)) Thus $PS \parallel AQ$ as the corresponding angles are equal 	2	2 marks • Correct proof 1 mark • Uses alternate segment theorem or similar in a valid approach
13 (b) (iii) $\therefore \angle PSA = \angle QAR$ (explained in (i)) $PA \parallel QR$ (similar method to part(ii) – commencing with $\angle TPS$) $\angle PAS = \angle QRA$ (corresponding $\angle$'s, $PA \parallel QR$) $\Delta PAS \parallel \Delta QRA$ (AA)	2	2 marks • Correct proof 1 mark • Uses properties of parallel lines or similar in a valid approach
13 (c) (i) $m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq}$ $= \frac{a(p+q)(p-q)}{2a(p-q)}$ $= \frac{p+q}{2}$ $y - ap^2 = \frac{(p+q)}{2}(x - 2ap)$ $2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$ $(p+q)x - 2y = 2apq$	2	2 marks • Correct proof 1 mark • Find the slope of the chord PQ
13 (c) (ii) focal chord passes through focus $(0, a)$ $(0, a) : (p+q)(0) - 2a = 2apq$ $-2a = 2apq$ $pq = -1$	1	1 mark • Correctly shows result
13 (c) (iii) Tangents and normals are perpendicular to each other $\therefore \angle NQT = \angle NPT = 90^\circ$ Thus $PTQN$ is a cyclic quadrilateral as a pair of opposite angles are supplementary	1	1 mark • Correctly explanation
13(c) (iv) $4ay = x^2$ $y = \frac{x^2}{4a}$ $\frac{dy}{dx} = \frac{x}{2a}$ when $x = 2ap$ $\frac{dy}{dx} = p$ Thus the slope of PT is p, and the slope of QT is q. As $pq = -1$, $PT \perp QT$ PQ is the diameter of the circle ($\angle$ in semicircle) Consequently C must be the midpoint of the diameter PQ	1	1 mark • Correctly explanation • Note: it is not necessary to derive the slope of the tangent.

Solution	Marks	Comments
13 (c) (v) $x_c = \frac{2ap + 2aq}{2}$ $= a(p + q) \Rightarrow p + q = \frac{x}{a}$ $y_c = \frac{ap^2 + aq^2}{2}$ $= \frac{a(p^2 + q^2)}{2}$ $= \frac{a[(p + q)^2 - 2pq]}{2}$ $= \frac{a}{2} \left[\left(\frac{x}{a} \right)^2 + 2 \right]$ as $pq = -1$ $= \frac{x^2}{2a} + a$ $\therefore$ locus is $y = \frac{x^2}{2a} + a$ OR $2a(y - a) = x^2$	2	2 marks • Correct solution 1 mark • Find the coordinates of C
QUESTION 14		
14 (a) (i) domain: $-1 \leq \frac{x}{2} \leq 1$ range: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$ $-2 \leq x \leq 2$ 	2	2 marks • Correct graph with domain and range indicated 1 mark • Identifies both domain and range • Correct curve
14 (a) (ii) $y = \sin^{-1}\left(\frac{x}{2}\right) \Rightarrow \sin y = \frac{x}{2}$ $x = 2 \sin y$  $\int_0^1 \sin^{-1}\left(\frac{x}{2}\right) dx = \frac{\pi}{6} \times 1 - 2 \int_0^{\frac{\pi}{6}} \sin y dy$ $= \frac{\pi}{6} + 2[\cos y]_0^{\frac{\pi}{6}}$ $= \frac{\pi}{6} + 2\left(\frac{\sqrt{3}}{2} - 1\right)$ $= \frac{\pi}{6} + \sqrt{3} - 2$	2	2 marks • Correct solution 1 mark • Correct answer obtained without referencing the graph • Uses the graph in a valid attempt to find the solution
14 (b) (i) $x = 4 \sin^2 t$ $\dot{x} = 8 \sin t \cos t$ $= 4 \sin 2t$ $\ddot{x} = 8 \cos 2t$ $= 8 - 16 \sin^2 t$ $= -4(4 \sin^2 t - 2)$ $= -4(x - 2)$ Hence the particle moves in SHM as $\ddot{x} = -4X$ where $X = x - 2$	2	2 marks • Correct solution 1 mark • Establishes a motion equation for acceleration • Recognises the condition for a particle to be moving in SHM
14 (b) (ii) maximum speed occurs at the centre of motion i.e. $x = 2$ $\dot{x} = 4 \sin 2t$ Thus the maximum speed is 4 m/s	2	2 marks • Correct solution 1 mark • Finds the centre of motion • Finds speed at the calculated centre of motion <i>Note: do not penalise for answers obtained from non-trivialised solutions to part (i)</i>

Solution		Marks	Comments
14 (c) (i)	When $t = 2$; $x = 8 \qquad y = -2$ $8 = 2V\cos\theta \qquad -2 = 2V\sin\theta - 19.6$ $V\cos\theta = 4 \qquad V\sin\theta = 8.8$ $\frac{V\sin\theta}{V\cos\theta} = \frac{8.8}{4}$ $\tan\theta = 2.2$	2	2 marks • Correct solution 1 mark • Uses $y = 4$ and concludes $\tan\theta = 2.95$
14 (c) (ii)	Projectile hits the ground when $y = -6$ $-6 = VT\sin\theta - 4.9T^2$ however $V\cos\theta = 4 \Rightarrow V = \frac{4}{\cos\theta}$ $-6 = \frac{4T\sin\theta}{\cos\theta} - 4.9T^2$ $4.9T^2 - 8.8T - 6 = 0$ $T = \frac{8.8 \pm \sqrt{8.8^2 + 4(6)(4.9)}}{9.8}$, but $T > 0$ $T = \frac{8.8 + \sqrt{195.04}}{9.8}$ $= 2.3230...$ ∴ the projectile hits the ground approximately 0.3 seconds after passing the pole	3	3 marks • Correct solution 2 marks • Finds the cartesian equation of motion that can be used to solve the problem, or equivalent merit 1 mark • Attempts to eliminate V from the motion equations • Finds a quadratic equation of motion.
14 (c) (iii)	$\dot{x} = V\cos\theta$ $= 4$ $\dot{y} = V\sin\theta - 9.8T$ $= 8.8 - 9.8(2.3)$ $= -13.74$ $\tan\alpha = \frac{\dot{y}}{\dot{x}}$ $= -\frac{13.74}{4}$ $\alpha = 73.76862255...$ ∴ the projectile makes an angle of 74° when striking the ground	2	2 marks • Correct solution 1 mark • Finds the horizontal and vertical components of velocity • Finds one component and uses $\frac{\dot{y}}{\dot{x}}$